Experiment 15 : Experiment to understand the data preprocessing in Data Science

Aim :

Experiment to understand the data preprocessing in Data Science

Description: Understand the importance of Data preprocessing in data science

Program Code:

```
import numpy as np
import pandas as pd

from google.colab import files

uploaded = files.upload()

df = pd.read_csv("/content/pre-process_datasample.csv")

print("Original Dataset:")
print(df)

print("\nDataset Information:")
df.info()

# Handle missing values
df["Country"] = df["Country"].fillna(df["Country"].mode()[0])
df["Age"] = df["Age"].fillna(df["Age"].median())
df["Salary"] = df["Salary"].fillna(round(df["Salary"].mean()))
```

```
print("\nDataset After Preprocessing:")
print(df)

# One-Hot Encoding
updated_dataset = pd.concat(
    [pd.get_dummies(df["Country"]), df.iloc[:, 1:]],
    axis=1
)

# Convert Purchased into numerical values
updated_dataset["Purchased"] = updated_dataset["Purchased"].replace(
    {"No": 0, "Yes": 1}
)

print("\nFinal Dataset:")
print(updated_dataset)
```

Output :

```
Original Dataset:
   Country   Age   Salary Purchased
0   France  44.0  72000.0        No
1    Spain  27.0  48000.0       Yes
2  Germany  30.0  54000.0        No
3    Spain  38.0  61000.0        No
4  Germany  40.0      NaN       Yes
5   France  35.0  58000.0       Yes
6    Spain   NaN  52000.0        No
7   France  48.0  79000.0       Yes
```

8 NaN 50.0 83000.0 No

9 France 37.0 67000.0 Yes

Dataset Information:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 10 entries, 0 to 9

Data columns (total 4 columns):

Column Non-Null Count Dtype

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0 Country 9 non-null object

1 Age 9 non-null float64

2 Salary 9 non-null float64

3 Purchased 10 non-null object

dtypes: float64(2), object(2)

memory usage: 452.0+ bytes

Dataset After Preprocessing:

Country Age Salary Purchased

0 France 44.0 72000.0 No

1 Spain 27.0 48000.0 Yes

2 Germany 30.0 54000.0 No

3 Spain 38.0 61000.0 No

4 Germany 40.0 63778.0 Yes

5 France 35.0 58000.0 Yes

6 Spain 38.0 52000.0 No

7 France 48.0 79000.0 Yes

8 France 50.0 83000.0 No

9 France 37.0 67000.0 Yes

Final Dataset:

	France	Germany	Spain	Age	Salary	Purchased
0	True	False	False	44.0	72000.0	0
1	False	False	True	27.0	48000.0	1
2	False	True	False	30.0	54000.0	0
3	False	False	True	38.0	61000.0	0
4	False	True	False	40.0	63778.0	1
5	True	False	False	35.0	58000.0	1
6	False	False	True	38.0	52000.0	0
7	True	False	False	48.0	79000.0	1
8	True	False	False	50.0	83000.0	0
9	True	False	False	37.0	67000.0	1

Result :

Thus, Experiment to understand the data preprocessing in Data

Science